

Techno-economic analysis of deuterium-tritium magnetic confinement fusion power plants

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HIGHLIGHTS

- Development of a bottom-up techno-economic analysis that provides a capital cost framework.
- Regulatory alignment is essential for Fusion's economic viability.
- Fusion-specific components and manufacturing costs are highly sensitive parameters.
- Fusion LCOE of \$140–\$550/MWh highlights its competitive potential.

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ABSTRACT

This techno-economic analysis of deuterium-tritium magnetic confinement fusion power plants (FPP) aims to assess the economic viability and scalability of FPPs in addressing global energy challenges and climate change. Amidst a backdrop of substantial investments in fusion technology, totaling \$7.1 billion to date, this study critically assesses the Nth-of-a-kind overnight capital costs of a FPP that hosts ARAI, a 350 MWe tokamak reactor based on the MIT ARC fusion concept. The overnight capital costs for ARAI-FPP are estimated to range between \$8800/kW and \$22,200/kW, with this variation largely driven by differing regulatory and manufacturing assumptions. The overall cost breakdown is found to be similar to past and recent fusion literature, where the direct cost of fusion reactor equipment is the largest cost driver. The levelized cost of electricity, which includes projections for operation and maintenance costs, are estimated to be between \$140/MWh and \$550/MWh. The findings aim to deepen the understanding of absolute and relative cost drivers in fusion energy and suggest strategies to improve its economic feasibility. The analysis underscores the crucial impact of fabrication costs and regulatory frameworks on cost dynamics, while also recognizing the substantial uncertainty in FPP cost estimates due to the low technology readiness level of several key components.

1. Introduction

As a major contributor to global CO₂ emissions, the United States (U.S.) actively engages in global initiatives to reduce its carbon footprint

and decarbonize its electric power systems to achieve net-zero carbon goals [1]. Energy use, through the burning of fossil fuels, is the largest driver of greenhouse emissions globally [2]. Renewable energy sources such as wind and solar are imperative for this transition but suffer from

Abbreviations: ARAI, Advanced Reactor Affordable Integration; ARAI-FPP, Advanced Reactor Affordable Integration Fusion Power Plant; ARC, Affordable Robust Compact; ARIES-AT, Advanced Research Innovation and Evaluation Study – Advanced Tokamak; ARIES-CS, Advanced Research Innovation and Evaluation Study – Compact Stellarator; ARIES-ST, Advanced Research Innovation and Evaluation Study – Spherical Torus; CF, Capacity Factor; COA, Code of Accounts; DOE, Department of Energy; D-T, Deuterium-Tritium; EEDB, Energy Economic Data Base; FOAK, First-of-a-kind; FPP, Fusion Power Plant; FPY, Full Power Years; HTS, High-Temperature Superconductor; IAEA, International Atomic Energy Agency; ITER, International Thermonuclear Experimental Reactor; kW, Kilowatt; LCOE, Levelized Cost of Electricity; MC, Magnetic Confinement; MIT, Massachusetts Institute of Technology; MWe, Megawatts Electric; MWh, Megawatt-hour; MWth, Megawatts Thermal; NCET, Nuclear Cost Estimation Tool; NRC, U.S. Nuclear Regulatory Commission; NOAK, Nth-of-a-Kind; NPP, Nuclear Fission Power Plant; OCC, Overnight Capital Costs; O&M, Operation and Maintenance; PWR12, 1200 MWe Westinghouse Pressurized Water Reactor; REBCO, Rare-earth barium copper oxide; R&D, Research and Development; TEA, Techno-Economic Analysis; TRL, Technology Readiness Level; U.S., United States; W, Watt.

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low power density, inefficient baseload performance, and a lack of energy storage solutions. Meanwhile, nuclear fission, while supplying a remarkable portion of the non-carbon-emitting energy in the world, faces high capital costs, cost overruns, and regulatory burdens, mainly due to long permitting processes and safety requirements for fission reactors [3]. These challenges prevent renewables and fission from displacing fossil fuels in the existing energy system. Fortunately, fusion energy provides a solution. Fusion energy mimics the sun's energy-producing processes and will generate virtually unlimited, clean energy. Its development may revolutionize power generation, altering our reliance on carbon-based fuels and facilitating a sustainable future in the U.S. and globally.

In 2022, the Biden-Harris Administration initiated a strategy to expedite the advancement of fusion power, setting a clear objective to exceed the break-even point (where the fusion system generates more energy than it consumes) [4]. This approach represents a substantial change from the 1970s, when most fusion research was predominantly confined to colleges and national laboratories with a focus on scientific demonstrations. Notably, in 2022, Lawrence Livermore National Lab surpassed the break-even point and successfully replicated the experiment three times, achieving a target gain of greater than one [5]. Today, this landscape has altered significantly due to the increase in private-sector investments and improved public-private partnerships, which have fostered meaningful progress toward the commercialization of fusion energy. To date, the fusion industry has accumulated over \$7.1 billion in investments; Commonwealth Fusion Systems has led the charge, raising over \$2 billion based on MIT's ARC concept, a Deuterium-Tritium (D-T) Magnetic Confinement (MC) reactor [6].

This analysis will present a methodology used to construct a techno-economic model for costing an Nth-of-a-kind (NOAK) Fusion Power Plant (FPP) with a focus on D-T MC technology. Therefore, both the presented lower bound and upper bound should be treated as NOAK. Additionally, this analysis addresses the impact of regulatory frameworks on capital costs, identifying how these requirements can contribute to cost escalations. As calculated by Stewart and Shirvan [7], the source of the regulatory-related escalations in labor rates, materials, and equipment mainly stems from requirements for quality assurance, control, and inspection. By integrating the latest technological innovations, including advanced materials, superconducting magnets, and advanced manufacturing capabilities, this analysis assesses how these innovations can reduce anticipated cost drivers. Moreover, this research will contribute its findings as a part of a new study by the MIT Energy Initiative aimed at assessing the role of fusion energy in a decarbonized electricity system [8].

According to the 2024 fusion companies survey by the Fusion Industry Association, the MC approach (51 %) represents the most adopted approach by the private Fusion Industry; similarly, the most pursued fuel source is D-T [6]. Moreover, D-T MC approaches have the most data available in the literature due to the extent to which MC reactors have been studied in universities and national laboratories. The D-T MC reactor technologies proposed by the ARIES studies and MIT's ARC conceptual design are parameterized for cost comparison [9–11]. As such, this analysis postulates a bottom-up capital cost methodology for a D-T MC FPP.

Despite the numerous publications on experimental FPP configurations, substantial uncertainty remains in cost estimates due to the absence of a demonstration FPP and a commercial supply chain for essential components. In order to partially address this uncertainty, the TEA work focuses on the NOAK cost and provides a cost estimation of FPPs given their proposed architectures. Estimating a NOAK cost avoids consideration of highly uncertain and sometimes controversial First-of-a-kind (FOAK) costs for initial demonstration power plants. Furthermore, various existing studies of FPPs, Fission, and other renewable energy sources will be leveraged to arrive at a range of possible capital and operating costs for FPPs. In particular, a discussion on the impact of regulation on the cost relative to a fission power plant will be made,

Table 1

General reactor parameters of D-T MC Reactor.

Parameters	Symbol	Base Case Values
Thermal Power (MWth)	P_{tot}	1000 MWth
Fusion Power (MW)	P_f	800 MW
Net Electric Output (MWe)	P_{net}	350 MWe
Efficiency (%)	η_{elec}	0.4
Capacity Factor	CF	0.7
Tritium Breeding Ratio	TBR	$1 \leq x \leq 1.1$

given the inherent merits of FPPs and lower overall radioactive inventory. Sensitivity to the identified cost drivers is also applied where deemed appropriate.

2. Methods

2.1. Code of accounts and scaling

Within this study, a cost accounting methodology was derived to calculate the Overnight Capital Costs (OCC), operation and maintenance costs (O&M), and Levelized Cost of Electricity (LCOE) for D-T MC Reactors. A Code of Accounts (COA) system used in nuclear fission power plant (NPP) cost estimates is from the U.S. Department of Energy (DOE) Economic Energy Data Base (EEDB). The COA has been accepted by the International Atomic Energy Agency, formalized by the Generation IV International Forum Economic Modeling Working Group [12] and adapted for some fusion applications [11]. A widely used cost accounting methodology facilitates uniformity and consistency when comparing the capital costs of NPPs across designs over time. The flexibility of the COA system allows for applications across nearly all nuclear power designs, which simplifies comparisons between NPP technologies. This flexibility explains why it has been used in many studies to estimate the costs of new power plants and compare costs across different nuclear and conventionally used power plant designs [13,14].

However, despite its flexibility and usage for comparisons across conventional power-generating facilities, the COA system itself was not designed for specific fusion concepts. As a result, the COA was modified to accommodate the differences between an FPP and NPP. These modifications are somewhat in line with previous adaptations of EEDB COA to fusion reactors [9–11]. The OCC within this TEA consists of Account 20: Direct Capital Costs, Account 30: Indirect Capital Costs, Account 40: Plant Decommissioning Cost, Account 60: Owner's Cost. Equation 2.1 depicts the OCC calculation:

$$OCC \left(\frac{\$}{kW} \right) = \frac{\text{Direct Costs} + \text{Indirect Costs} + \text{Decommissioning Cost} + \text{Owner's Cost}}{P_{net} \times 1,000} \quad (2.1)$$

2.1.1. Concepts and assumptions

There are several assumptions affecting the FPP costs, its supporting equipment, and the facilities that house it. Table 1 depicts relevant assumptions regarding the D-T MC FPP considered in this study. Many features of the MC reactor discussed within the analysis are based on the leading candidate materials shown in the annual Fusion Industry Association report, where companies self-report their progress [15]. The assumption behind the tritium breeding ratio results in assuming no fuel capital cost for this NOAK analysis. This analysis also acknowledges the amount of uncertainty within the system and highlights the need for technological advancements to advance economically critical features.

The following are noteworthy assumptions in the TEA:

- Publicly available data was used for the costs of materials involved in the construction and annual operation of a FPP. Therefore, it cannot, by definition, capture recent improvements in science or manufacturing held by private companies, yet will still provide a relative basis from which to assess fusion economics.

Table 2

Direct costs.

COA Number and Name	Description
20 Direct Costs	
21.1 Land & Land Rights	
21.2 Structures & Site Facilities	<i>Described in Section 3.2</i>
22 Reactor Plant Equipment	Vacuum vessel, Blanket, Magnet, Fuel Handling System, Tritium Extraction and Removal, Auxiliary Cooling Systems, Cryostat, Other Reactor Plant Equipment (e.g., divertor), Instrumentation and Control
23 Turbine Generator Equipment	Rankine Cycle assumed: Turbine Generator, Condensing Systems, Feed Heating System, Other Turbine Plant Equipment
24 Electric Plant Equipment	Switchgear, Power and Control Wiring, Switchboards, Station Service Equipment, Electrical Structures and Wiring Containers, Electrical Lighting
26 Heat Rejection System	Heat exchangers and steam generators for steam production. Pumps, piping, valves required for heat removal.
27 Misc. Plant Equipment	Costs for transportation and installation of the reactor plant equipment

- This TEA analysis does not reflect the business plans of any particular fusion company. Instead, concepts will be explored based on their popularity.
- The fusion industry requires significant novel technologies and concepts to be developed in order to ensure economically feasible deployment, and this analysis does not aim to capture the future state of fusion R&D.
- All civil, mechanical, and electrical work are costed based on a thermal plant that has a 1000 MWth and a 350 MWe electrical power output.
- Fusion-specific technology possesses great uncertainty, so assumptions about materials needed were based on experimental fusion concepts and the Fusion Industry Association report.

2.2. Direct costs

The Direct Capital Costs (shown in Table 2) were determined through bottom-up and top-down assessments. While bottom-up assessments provide a robust estimate by considering fabrication, material inputs, and site labor, top-down assessments were employed when assessing bottom-up quantities was challenging. The bottom-up estimate employs a combination of methods to compute a range of capital costs, and the Nuclear Cost Estimation Tool (NCET) was referenced because it provides a bottom-up cost estimation tool for fission NPPs [14].

Within this analysis, the scaling methodology from NCET, was utilized and modified to gauge the costs associated with a FPP. NCET has been used to model ~45–1500 MWe fission modular power plants from FOAK to NOAK. Similar to fission, nuclear fusion demonstrates a high initial capital cost with specialized equipment while benefiting from low fuel costs by leveraging the high energy density of the nuclear reactions. The cost estimate tool within NCET utilizes scaling methods for direct costs from the EEDB PWR12, a 1200 MWe pressurized water reactor, but for this TEA, the “median experience” costs were not used, instead the lower “better experience” costs were used to derive the scaled values used in this TEA. As it will be shown, the “better experience” costs for the non-nuclear part of the NPP are similar to the costs for NOAK thermal power plants (e.g., combined cycle natural gas) constructed today, justifying its utilization for this NOAK focused TEA. Equation 2.2 was used for scaling:

$$C = C_{EEDB} * \left(\frac{P_{new}}{P_{EEDB}} \right)^n \quad (2.2)$$

P_{new} represents the value of a new parameter such as superstructure

volume, superstructure area, building volume, substructure area, etc. P_{EEDB} represents the reference parameter from the EEDB. C_{EEDB} represents the reference cost in the EEDB. The exponent n represents a scaling factor which depends on each structure, system and component. Lastly, C represents the new cost of the scaled component is then taken to and divided by the reference EEDB cost. Given that an FPP has never been constructed, no historical data exists to validate the fusion-specific cost numbers. Thus, this methodology will be used to examine the *relative* economics of a D-T MC FPP. PWR12 costs include factory costs, site labor hours, material costs, and quantity. The direct costs are comprised of 7 top-level accounts and 27 subaccounts. The top-level direct accounts are structures and improvements (representing the civil works), reactor plant equipment (will be replaced with FPP specific equipment), turbine generator equipment (assumes steam Rankine cycle similar to FPP), electrical plant equipment, miscellaneous equipment, and the main heat rejection system.

2.2.1. D-T concepts

Given that MC concepts have been researched for decades, there is extensive literature describing various MC designs. Advanced Reactor Innovation and Evaluation Study (ARIES) presented several capital cost studies for various MC concepts which will be referenced in this analysis, including the Spherical Torus (ARIES-ST) and Compact Stellarator (ARIES-CS). The costs of those concepts will be discussed in the results section. Affordable, Robust, Compact reactor (ARC), piloted out of MIT, utilizes rare-earth barium copper oxide (REBCO) high-temperature superconductors (HTS), and within the original ARC study, a FOAK cost for 70kA REBCO cables was provided [16]. All ARIES concepts were intended to produce 1000 MWe of net electric power. The ARIES-ST power core weighs approximately 8613 t, and the ARIES-CS power core weighs roughly 12,555 t [17,18]. Meanwhile, ARC was designed to produce 190 MWe of net electric power, and its power core weighs a total of 7190 t. Therefore, ARC features a significantly higher mass-to-power ratio compared to the ARIES reactor series. Given HTS's impacts and continued presence in the fusion industry, this TEA modifies the ARC concept as a basis for the bottom-up estimate. Modifications were made to the ARC concept to apply leading candidate materials, such as vanadium alloys, to the design of the replaceable components, including the vacuum vessel and plasma-facing components. Within ARC, a total of 5730 km of 70kA REBCO cables were utilized, and thus, the same amount will be assumed within this analysis. Each cable consists of a stack of REBCO tapes, each tape measuring 12 mm in width and 0.1 mm in thickness. Again, these modifications were based on the leading candidate materials discussed in the Fusion Industry Supply Chain report [15]. The modified ARC concept will be labeled as the Advanced Reactor Affordable Integration (ARAI). ARAI will be compared to ARIES-ST and ARIES-CS to demonstrate the flexibility and inclusivity of the bottom-up costing approach.

Tokamak designs researched in the U.S., first studied at the University of Wisconsin–Madison, were considerably larger than ARAI; for example, UWMAK I & II (1970s) had projected masses of approximately 61,000 t [19]. From the outset in the 1970s, funding priorities and research directions in U.S. fusion programs emphasized cost-effectiveness as a central objective in fusion reactor design—one of the motivating pillars of ARIES [20]. Large reactors were widely viewed as capital-intensive, which motivated the pursuit of technologies such as HTS. HTS allowed for stronger magnetic fields and, ultimately, a key requirement of compact devices, such as the ARC concept.

The International Thermonuclear Experimental Reactor (ITER), a major D-T MC fusion research and engineering project, has experienced delays and cost overruns throughout the course of its development. The U.S. halted its participation in 1998 due to concerns over the program's cost, rejoining in 2003 after more technical and economic refinements were introduced [21]. ITER's technical and economic trajectory should not be considered representative of what a FOAK fusion power plant would cost, as ITER was not designed to be a power plant and it faced

Table 3

Components of ARAI.

Account Number and Name	Tonnes	Material assumed
22.11 Vacuum vessel	201	
22.11.1 First wall	3.72	Tungsten
22.11.2 Inner vacuum vessel wall	100	V-4Cr-4-Ti
22.11.3 Outer vacuum vessel wall	86.4	V-4Cr-4-Ti
22.11.4 Vacuum vessel ribbing	6.8	V-4Cr-4-Ti
22.11.5 Vacuum vessel posts	4.14	V-4Cr-4-Ti
22.11.6 Vacuum Pumps		
22.12 Blanket	731.5	
22.12.1 Blanket tank	97.1	SS316 LN
22.12.2 TiH ₂ shield	26.4	
22.12.3 Channel FLiBe	8.1	FLiBe
22.12.4 Blanket tank FLiBe	300	FLiBe
22.12.5 Heat exchanger FLiBe	300	FLiBe
22.13 Magnet	3358	
22.13.1 Magnet structure	3000	SS316 LN
22.13.2 REBCO tape structural materials	358	Copper

Table 4

Material cost assumptions.

Material Type	Material Cost
V-4Cr-4-Ti	\$37/kg
SS316 LN	\$10/kg
FLiBe	\$154/kg
Tungsten	\$29/kg
TiH ₂	\$26/kg
Copper	\$8.3/kg

challenges in its development partly due to its structure as a multinational collaboration with fluctuations in its funding and resources. However, ITER demonstrates a history of marrying technical and economic considerations that have parallels to the history of and expectations around the development of cost-effective fusion, and this ultimately, reflects a broader commitment within the U.S. D-T MC fusion community to build fusion energy with cost-effectiveness in mind.

2.2.2. Reactor plant equipment

The reactor plant equipment (Account 22) will harbor the power core of ARAI and the majority of the fusion-specific technology. ARAI will employ the following materials for its assembly. The blanket tank and magnet structure will be crafted from stainless steel SS316 LN. Enclosing the blanket tank, a neutron shield composed of titanium dihydride (TiH₂) will be designated as the shield. The vacuum vessel will utilize the vanadium alloy V-4Cr-4-Ti, and the first wall will be composed of tungsten tiles. Additionally, the blanket is molten salt fluorine, lithium, and beryllium (FLiBe). Table 3 lists the mass of these components. The power core of ARAI, composed of the vacuum vessel, blanket, and magnet, weighs 4291 t.

There exists great uncertainty pertaining to the cost of FLiBe, vanadium alloys, tritium, REBCO, and other specialized materials needed for the power core and for operation. In addition, due to the immature market of these materials, there is an absence of publicly accessible costing data. Hence why the material costs were obtained from a diversity of sources. A FOAK cost of FLiBe was taken from an internal source, and its NOAK cost was derived by applying a 20 % learning rate. The raw material costs of most of the components except the vacuum vessel were obtained from Sorbom et al. but not scaled to \$2021, and those costs are comprised of both historical data and private quotes obtained from vendors (Table 4) [16]. For the vacuum vessel material, V-4Cr-4-Ti was created by obtaining material costs of Vanadium, Chromium, and Titanium and costing the V-4Cr-4-Ti based on its composition [22–24]. After costing the alloy based on composition, a 15 % premium was added for the removal of impurities, resulting in a cost of \$37/kg (Table 4). It should be noted that material costs are a sensitive

Table 5

Additional reactor plant equipment cost accounts.

22.14	Supplemental Heating Systems
22.16	Power Supplies
22.18	Divertor
22.2	Main Heat Transfer & Transport Systems
22.4	Radioactive Waste Treatment & Disposal
22.5	Fuel Handling System
22.6	Other Reactor Plant Equipment
22.7	Cryosystem

parameter, and impacts on it inevitably will affect the total cost of the lifetime and replaceable components. Table 4 depicts the material costs utilized.

In addition to the vacuum vessel, blanket, and magnet system costs, the reactor plant equipment includes costs of the vacuum systems (cryopumps), cryosystem, divertor, supplemental heating/current drive systems, tritium handling, and heat exchangers (Table 5).

Cost assumptions for the vacuum systems include \$0.7 million for 48 cryopumps, based on commercial rates. The supplemental heating/current drive systems and power supplies are estimated to cost about \$2.5/W and \$1.5/W, respectively. The costs for the divertor include the purchase of 18 tungsten tiles for \$150 million, while the cryosystem costs account for both material and fabrication expenses, alongside a fixed cost per kW for cryocooling at 20 K, totaling approximately \$300/kW.

Other cost estimates within the reactor plant equipment are derived using the NCET tool, with adaptations for technologies like tritium handling and heat exchangers based on fission technology costs. Cost adjustments for magnets from the ARC design incorporate learning rates and manufacturing complexity, differentiating between FOAK and NOAK assumptions. These cost details and comparisons with ARIES-ST and ARIES-CS are discussed in Section 4.3. Additionally, while some of these technologies—such as tritium handling systems, advanced heat exhaust solutions, and high-field HTS magnets—currently possess relatively low Technology Readiness Levels (TRL), this analysis assumes a NOAK deployment scenario, where present TRL limitations are not treated as constraints.

2.2.3. Structures and site facilities

Utilizing NCET scaling methods and the ITER facility layout, this analysis identified and scaled necessary buildings for MC fusion designs. Structures and Site Facilities (Account 21.2) emerge as a significant cost driver, mainly due to safety regulations. Unlike fission plants, U.S. fusion plants might face simpler regulations, suggesting a potential for cost reductions. Regulatory impacts typically double building costs by a factor of about 2.2 [14]. This analysis, therefore, adjusts the capital cost estimates for reactor buildings by this factor, considering the potential differences in regulatory impacts between fission and fusion.

Detailed architectural plans from ITER were used to adjust the size and design of containment buildings for compact MC reactors using HTS, which require smaller facilities compared to those using low-temperature superconductors. Calculations of material volumes and surface areas—such as concrete and steel—provided the basis for bottom-up cost estimates of various building sections, including the control room and waste processing facilities. Additionally, the functional equivalence of turbine generators with those used in existing nuclear and conventional power systems was assumed.

2.2.4. Conventional equipment assumptions

When beginning a bottom-up estimate, it is important to distinguish NOAK assumptions relevant to existing technology and processes with available cost data. Thus, several assumptions in this analysis rely on cost data for conventional thermal plants. Turbine Generator Equipment (Account 23), Electric Plant Equipment (Account 24), and Heat rejection system (Account 26) reflect components used in conventional thermal

Table 6
Indirect cost formulas.

30	Indirect Costs	Formula Used
31	Design Services at Home Office	$0.14 \times \text{Direct Costs}$
32	Project & Construction Management at Home Office	$0.01 \times \text{Direct Costs}$
34	Field Construction Management at Plant Site	$0.01 \times \text{Direct Costs}$
35	Field Construction Supervision at Plant Site	$0.05 \times \text{Direct Costs}$
36	Field Indirect Costs	$0.16 \times \text{Direct Costs}$
37	Plant Commissioning Service	$0.01 \times \text{Direct Costs}$

plants and, therefore, possess NOAK costs. These assumptions can also be used to estimate the savings that can be achieved through brownfield siting. By pursuing brownfield siting, existing facilities can be used and, therefore, will not need to be repurchased or reconstructed, so the sum of the costs in Accounts 23, 24 and 26 gives a rough approximation of how much can be saved by not having to re-purchase those components. This provides an upper bound for brownfield savings since it is likely that only the switchyard and parts of the heat rejection system can be fully reused.

2.3. Indirect costs

The top-level indirect costs are construction services, engineering and home office services, and field supervision and offsite services, as listed in Table 6. The Indirect Costs were estimated as a percentage of direct costs. The percentages given are based on Gen 4 fission data sets that approximate indirect costs to reflect the experience of the design and construction techniques [25]. The indirect cost will depend on ranges for direct cost, learning rate, modularization (percent of work onsite versus offsite), and standardization of design. The indirect costs have great potential to be a cost driver within the analysis, as experience and regulatory constraints can impact indirect costs. In many FPP cost studies noted in the literature, indirect costs align with direct costs [10]. For fission power plants, indirect costs as a percentage of the total cost can be as much as ~80 % of the cost [26] while for natural gas plants are about 20 % of the total cost [27]. In the EEDB database, the better experience plant realizes about 40 % indirect cost as a portion of the total cost. In this TEA, the indirect cost was assumed to be 20 % and 40 % of the total cost, for the lower and upper bound cost, respectively, since a NOAK FPP will likely lie between a natural gas plant and an advanced fission plant.

2.4. Operation and maintenance costs

Different reactor designs necessitate different maintenance approaches, horizontal or vertical integrated maintenance strategies being prominent contenders. Within the reactor building, the bottom-up estimate assumes a vertical maintenance approach where the reactor vessel would be lifted by a crane to perform certain maintenance tasks. The reactor building layout allows for remote O&M operations through a combination of vertical and port-based maintenance. An optimized maintenance schedule would include planned and unplanned maintenance timeframes for the reactor and auxiliary plant systems. Again,

Table 7
O&M cost assumptions.

Account #	Category	Lower Bound	Upper bound
71	Fixed O&M:	50 Employees	95 Employees
	FTE Costs (Assuming each employee has a Salary of \$200,000)		
72	Annual Variable O&M	$\sum (\text{Annual Equipment Maintenance} + \text{Power Core Replaceable Components})$	
72.1	Annual Equipment Maintenance (without Replaceable Components)	$3\% \times \{\text{Reactor Plant Equipment (not including replaceable components)} + \text{Turbine Generator Equipment Cost} + \text{Electric Plant Equipment Cost}\}$	
72.2	Annual Power Core Replaceable Components	$\left(\frac{\text{Cost of Replaceable Components}}{\text{Frequency of Replacement}} \right) \times (\text{Transportation Costs} + \text{Installation Costs})$	

given the amount of uncertainty in the maintenance schemes and material lifetimes, we are challenged to estimate an annual O&M cost based on the cost of processes. Thus, for this analysis, O&M costs are based on a percentage of costs in the Reactor Plant Equipment, Turbine Generator Equipment, and Electrical Plant Equipment. The modularity of replaceable components is critical to efficient maintenance. Ultimately, the maintenance of power core components needs to be efficient and timely to ensure that maintenance downtimes are minimized. A great amount of R&D is required in order to have a remote handling system capable of effectively and efficiently evaluating, diagnosing, removing, and installing power core components. The O&M costs have three categories: Fixed O&M, Variable O&M, and Replaceable Components cost.

Fixed O&M (Account 71) costs pertain to the number of full-time-employees (FTE) expected to be at a NOAK FPP. Numbers from other industries were utilized to estimate the number of FTEs at a NOAK FPP. For instance, a thermal solar plant, whose power production system closely resembles D-T fusion energy systems (molten salt storage with superheated power cycle), currently employs 85 FTEs at a site in the U. S., which will be used as the lower bound assumption. An advanced nuclear fission plant has been noted to potentially require 200 FTEs, half of which are dedicated to security forces [14]. 95 FTEs are assumed for the upper bound in recognition of the added complexity of operating a fusion power plant compared to a solar thermal (Table 7). As the lower bound, approximately half that number is assumed based on a scenario in which multiple reactors are installed at the same site to enable the staff to efficiently support multiple plants (Table 7).

Annual Variable O&M (Account 72) represents the sum of the Annual Equipment Maintenance and the Replaceable Components Costs. Annual Equipment Maintenance consists of roughly 3 % of the costs associated with Reactor Plant Equipment (not including replaceable Fusion reactor components), Turbine Generator Equipment, and Electric Plant Equipment without including transportation and installation costs (Table 5).

Replaceable Component costs and frequency of replacement will depend on the specific design of the FPP (Table 7). Choices such as fusion fuel, containment method, material of construction, and neutron flux for each component are key differences. D-T compact FPP concepts will have more frequent first wall component replacement rates than aneutronic FPP concepts. In ARAI, the replaceable components of the reactor core include the vacuum vessel and plasma-facing components.

2.5. Decommissioning and owner's cost

Decommissioning costs (Account 40) were calculated as 3 % of the total capital costs. It should be noted that the amount of solid waste created by the disposal of replaceable components is an important topic in terms of both worker health and public safety [28]. The decommissioning of FPPs is an area that has not been significantly researched, but lessons can be learned from the fission industry.

Owner's Cost (Account 60) is bundled with typical contingency costs and computed by multiplying the direct cost lower bound by 25 %, which results in the owner's cost representing ~17 % of the overnight capital costs in the lower bound case. This amount is similar to a typical thermal power plant assumption. For NPPs, this value is closer to 45 %

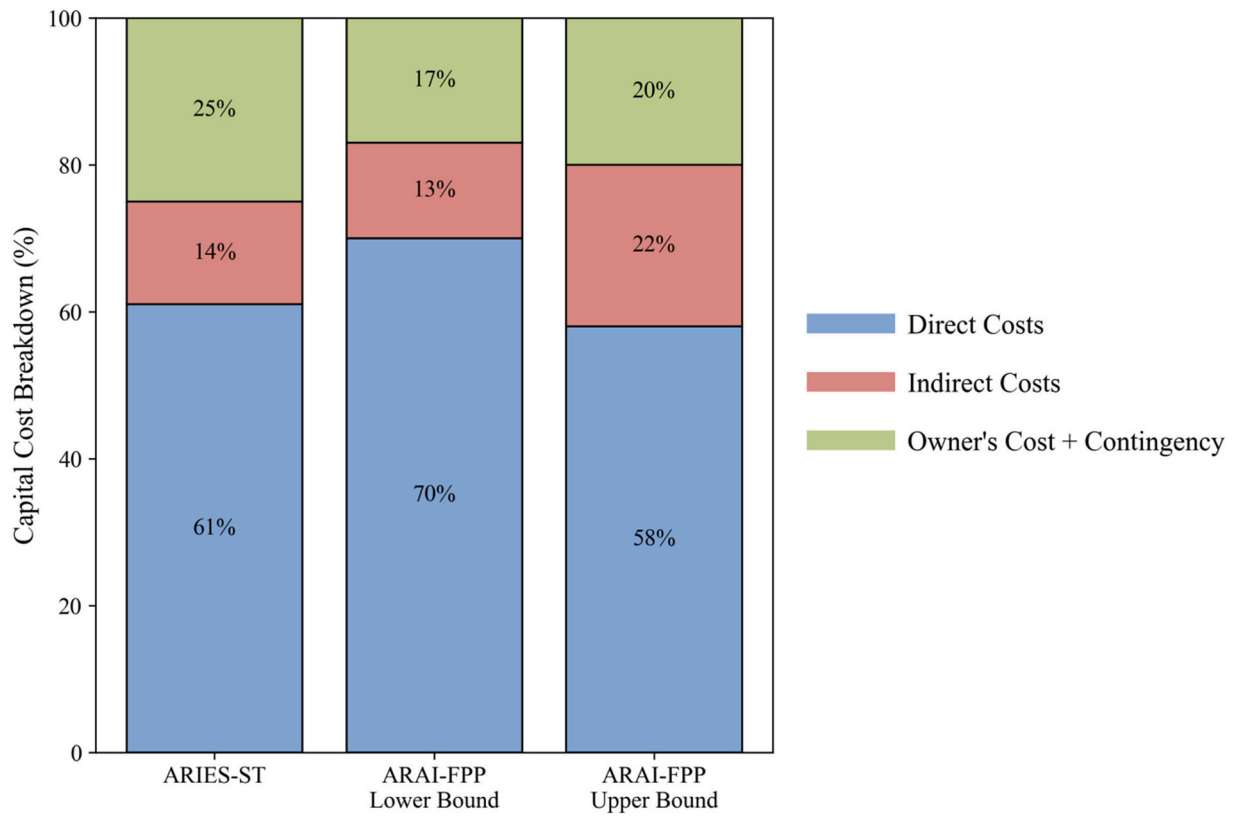


Fig. 1. High-level capital cost breakdown for the ARIES-ST from literature compared to ARAI-FPP lower and upper bound.

[29], for the upper bound, the direct cost was multiplied by 35 % to capture limited nuclear safety escalation.

2.6. Capacity factor and levelized cost of electricity

2.6.1. Capacity factor

By 2023, water-cooled NPPs achieved a capacity factor (CF) over 90 % for the last decade, significantly higher than offshore wind and solar PV, which did not exceed 45 % and 18 % respectively [30,31]. This positions nuclear as a promising low-carbon option. Initially, NPPs struggled with high CFs, but adaptation and significant operational costs drove improvements.

In this TEA, NOAK fusion plants' CF ranges from 0.5 to 0.7. The upper CF reflects non-water-cooled NPPs and fossil plants, targeted replacements in the energy grid [32]. The lower CF incorporates a 0.2 penalty for additional challenges, informing the LCOE.

FOAK steady-state tokamaks need to address high recirculated power and low CF to boost efficiency. Conversely, NOAK devices should achieve low recirculated power and high CF for economic competitiveness [33].

2.6.2. Levelized cost of electricity

The LCOE is characterized as the price at which the generated electricity should be sold for the system to break even at the end of its lifetime [34]. A simplified LCOE calculation was taken from National Renewable Energy Laboratory [35] and adapted for an equation that calculates an LCOE based on the following inputs:

- r = discount rate per year (6 %)
- N = total plant lifetime in years (30 years)
- Capacity Factor = between 0.5 and 0.7
- Hours in a year = 8760 h
- Annual O&M $\left(\frac{\$}{\text{MWh}}\right)$

Table 8

Direct plus Indirect cost comparison for the ARIES-ST, ARIES-CS, and ARAI-FPP lower and upper bound.

Magnetic Confinement Concept	Direct + Indirect Cost
ARIES-ST	\$9800/kW
ARIES-CS	\$8200/kW
ARAI-FPP Lower Bound	\$7100/kW
ARAI-FPP Upper Bound	\$18,500/kW

$$\bullet \text{ OCC } \left(\frac{\$}{\text{kW}} \right)$$

$$\text{Capital Recovery Factor (CRF)} = \frac{[r \times (1+r)]^N}{(1+r)^N - 1}$$

$$\text{LCOE} \left(\frac{\$}{\text{MW}} \right) = \left(\frac{\text{OCC} \times \text{CRF}}{\text{CF} \times \text{Hours in a year}} \times 1,000 \right) + (\text{Annual O\&M}) \quad (2.3)$$

3. Results

3.1. Overnight capital cost results

Fig. 1 displays a comparison of the direct cost breakdown amongst the ARAI-FPP lower and upper bound, ARIES-ST. The ARAI-FPP upper and lower bounds show similar breakdowns of direct, indirect, and owner's cost plus contingency to literature findings. These comparisons demonstrate areas of relative importance provided in the fusion literature.

3.2. Direct costs

In the MC cases noted, the direct cost constitutes most of the OCC, as

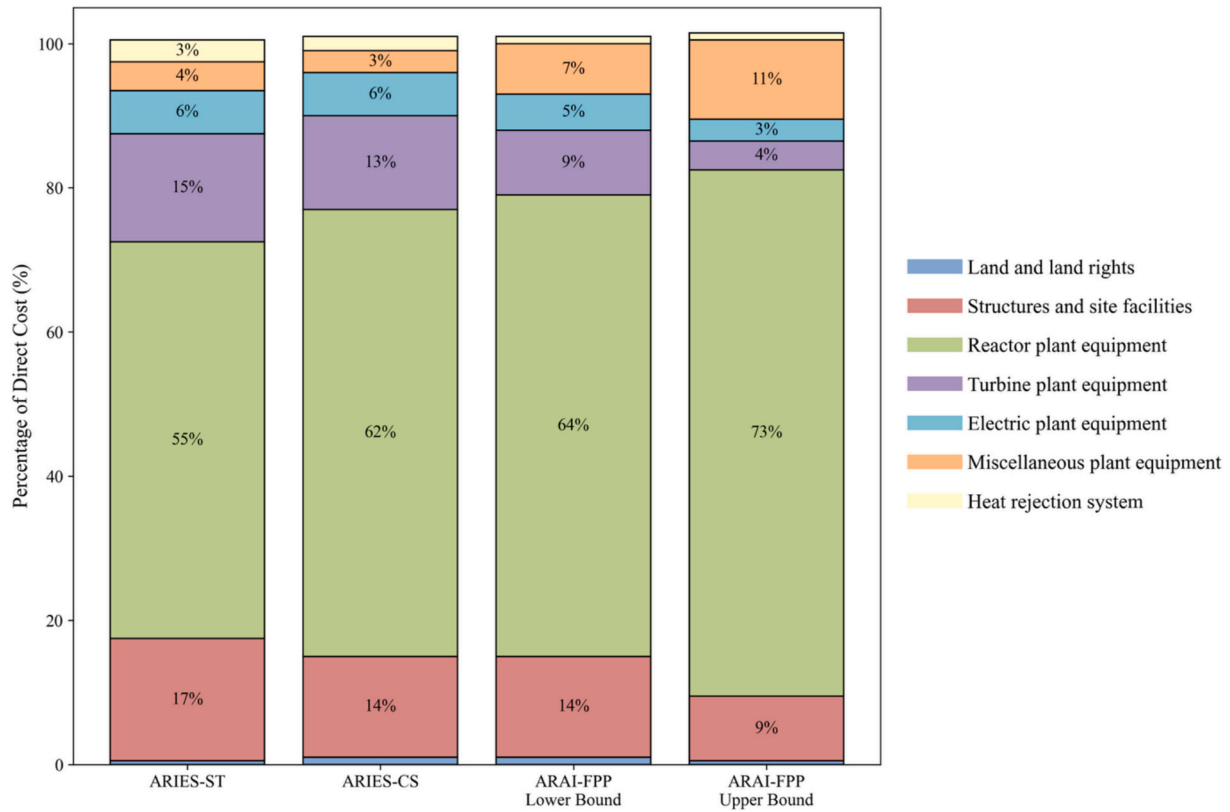


Fig. 2. Direct Cost Breakdown Comparison amongst lower and upper bound of ARAI-FPP, ARIES-ST, and ARIES-CS. Percentages less than 3 % are excluded from labels to maintain readability.

Table 9

O&M, decommissioning and owner's costs.

Account #	Account Name	Lower Bound	Upper Bound
71	Fixed annual O&M	\$5/MWh	\$12/MWh
72	Variable general O&M	\$30/MWh	\$170/MWh
72.1	Annual Equipment Maintenance (without Replaceable Components)	\$19/MWh	\$63/MWh
72.2	Annual Power Core Replaceable Components (Including Transport and Installation)	\$11/MWh	\$107/MWh
40	Plant Decommissioning Cost	\$64.4 M	\$96.6 M
60	Owner's Cost	\$1501/kW	\$3260/kW

shown in Fig. 1. Table 8 compares the total direct cost for the MC concepts divided by the respective kW power output assumption. Fig. 2 shows the relative breakdown of this direct cost. These results ignore the potential of each approach in reaching viability from the point of view of physics and engineering. The lower bound direct cost estimate for ARAI-FPP aligns with the literature values of ARIES-ST and ARIES-CS. This consistency and similarity amongst the lower bound cost are expected since the ARIES costing analysis and the literature both assumed a conventional manufacturing cost per kg of material input for the cost analysis.

The upper bound ARAI-FPP is nearly 3× the cost of the lower bound and about double the cost of the ARIES-ST. This is due to the differences in direct and indirect cost assumptions. To better understand this source of difference, it is necessary to further break down the direct costs to understand the origin of these costs. To facilitate a similar comparison between the ARIES-CS, ARIES-ST, and ARAI-FPP, the direct cost accounts have been modified to reflect a similar structure to the ARAI-FPP. As shown, the direct cost accounts of ARAI-FPP align with the ARIES-ST and ARIES-CS. The cost drivers of the analysis primarily reside in the reactor plant equipment. Thus, a cost driver analysis was conducted to further understand the cost driver within the reactor plant equipment.

3.3. Operations and maintenance, decommissioning and owner's cost

Section 2.4 explains how a range of potential O&M costs for ARAI-FPP were calculated. The O&M costs for the ARIES reactors were estimated to be lower than ARAI-FPP since the vacuum vessel in those designs have low neutron fluxes relative to the ARAI-FPP concept. Most of the existing literature, including ARIES, quotes O&M cost in the range of \$20–30/MWh when inflated to \$2021, they are line with the lower bound for ARAI-FPP. The Decommissioning and Owner's Costs were calculated based on the methodology discussed in Section 2.5. As shown in Fig. 1, the owner's cost assumption of the ARAI-FPP is in line with the costs posed in the ARIES studies. Table 9 shows the lower and upper bound Decommissioning and Owner's Costs.

3.4. Levelized cost of electricity

Comparing the LCOE of other energy generation technologies through an unsubsidized cost analysis provides a more suitable point of comparison for the cost of a FPP to both renewable and conventional energy technologies. Fig. 3 depicts the lower and upper bound LCOE of the ARAI-FPP alongside the LCOEs computed through using

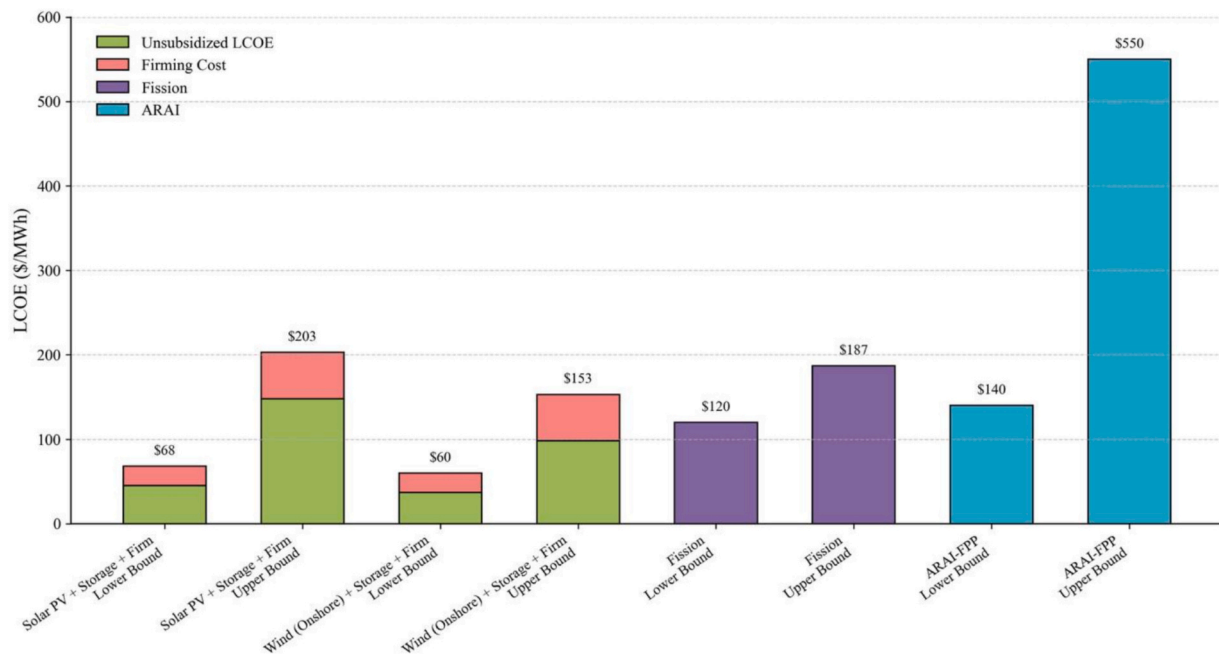


Fig. 3. LCOE Comparison with Lazard 2024 unsubsidized input data and firming cost for renewables using eq. 2.3 compared to NOAK ARAI LCOE estimates.

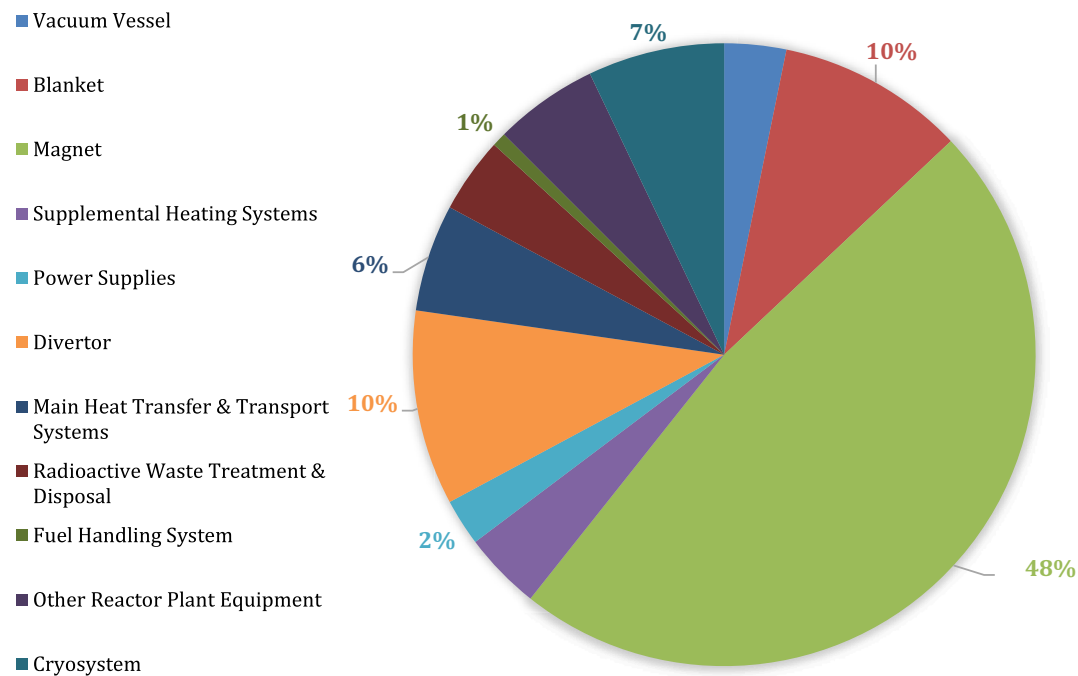


Fig. 4. Reactor plant equipment breakdown.

unsubsidized cost data by Lazard 2024 report as input to eq. 2.3. The cost of Wind (Onshore) and Solar PV reflect the cost of firming their intermittency and storage [36]. As shown, the NOAK lower bound for the considered FPP can be competitive with the upper bound of other clean sources of electricity. Meanwhile, the upper bound cost is not competitive with the other sources of electricity, which motivates further exploration of cost reduction strategies for the FPP, including consideration of other less mature fusion concepts such as magnetoinertial. The costs for Solar PV and Wind (Onshore) are based on Lazard 2024's estimates for the CAISO region. In CAISO, Lazard's total LCOE for renewables includes both generation and firming costs, with

firming based on the Net Cost of New Entry for a 4-h lithium-ion battery system. Firming needs are determined using Effective Load-Carrying Capability, which is relatively low for wind (often ~15 %), requiring most of its capacity to be supplemented by additional firm resources to meet reliability targets. For PV + Storage systems, Lazard assumes 50 % of the PV nameplate capacity is paired with 4-h storage, and only the portion that can be charged directly by solar and discharged during peak demand is counted as firm—resulting in a total cost that exceeds unsubsidized LCOE values alone. Lazard's LCOE estimate for conventional nuclear energy ranges are based on assumptions pertaining to Vogtle Plant Units 3 and 4. These values reflect high capital costs (~\$31.5

billion for ~ 2.2 GW), a 97 % capacity factor, and an assumed plant lifetime of 60–80 years. The financial assumptions—60 % debt at 8 % interest and 40 % equity at 12 % return—are consistent across technologies. It should be noted that Lazard's data for the other sources may not represent true NOAK costs as well as Lazard's firming cost which is derived from using combined cycle natural gas peakers, making the renewable energy sources not truly zero emission [36].

4. Discussion

4.1. Cost driver analysis

This section delves into the cost drivers outlined by the ARAI-FPP. A more in-depth examination of these cost drivers is essential, given that our current cost estimates for the tokamak are based on a nascent understanding of the technology and processes required to build FPPs. Outside of the potential cost escalation borne by regulatory requirements, the most significant capital cost components within MC concepts are related to the reactor plant equipment.

4.2. Overview and reactor plant equipment breakdown

To start, the reactor plant equipment, the cost driver of the OCC, will be discussed, as shown in Fig. 4.

The power core comprises the vacuum vessel, blanket, magnet system (toroidal field coils, poloidal field coils, auxiliary coils, superconductors), and divertor. The power core components are the cost drivers of the reactor plant equipment. However, upon further investigation, it is apparent that most of the costs of the power components pertain to the manufacturing and fabrication of raw materials used in the components mentioned.

The magnet system represents ~ 48 % of the Reactor Plant Equipment costs (Fig. 4). While tokamaks have transitioned to adopting smaller devices, thus reducing the mass of magnet structures per fusion power produced by more than 50 % compared to ITER, magnet components remain significant cost drivers. The magnet structure, unlike other reactor components, has established supply chains for its core material, and its raw material cost is comparatively lower than the raw material costs for the vacuum vessel and blanket. The difference between the fabricated cost of a component and its material cost is notable. In the 2015 ARC paper, the fabrication costs were estimated to be a factor of $20\times$ greater than the raw material cost [16]. Thus, in addition to providing an overview of the ARC fabrication costing, the deduction of the FOAK cost will be made and compared with NOAK fabrication cost assumptions. Given that Fusion's deployment of the NOAK plant is expected between the 2040 to 2050 timeframe, it is important to understand how advanced technologies can alter standardized manufacturing processes and costs. Thus, there will be a discussion about different cost assumptions, such as alternative low-cost fabrication methods through different manufacturing approaches and the costs of mass-manufactured components. Finally, this section will explore alternative costing methods to understand the impacts on the overall capital cost by applying a new lower bound to the fabrication costs. But first, it is important to understand the fabrication costs included within fusion literature.

4.3. Assumptions in fabrication cost estimation and FOAK fabrication costs

Existing fusion economics models often lack clarity regarding the assumptions underpinning the cost of component fabrication. Additionally, existing simplified fabrication cost methods can bring large uncertainty in fabrication cost estimates because they are nonspecific to reactor components or materials [37]. Highly referenced sources such as ARC: A compact, high-field, fusion nuclear science facility and demonstration power plant with demountable magnets utilize these simplified

Wright's Law: FOAK Fab. Cost to NOAK Fab. Cost

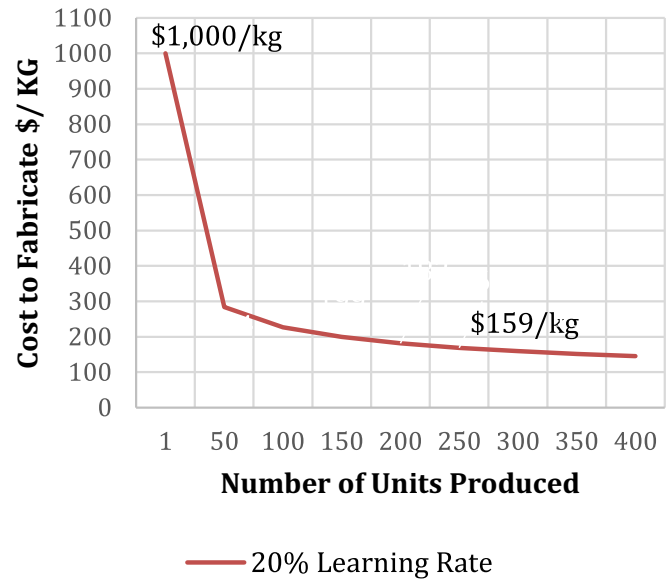


Fig. 5. FOAK to NOAK fabrication cost reduction vs. number of FPPs.

Table 10

Cost of fabricated power components.

Fabricated Power Core Components	Lower bound
Fabricated Magnet Cost (SS316 LN)	\$1371/kW
Fabricated Blanket Tank Cost (TiH2)	\$63/kW
Fabricated VV Cost (V-4Cr-4-Ti) + First Wall (Tungsten)	\$109/kW
HTS Fabricated Costs	\$162/kW
Lifetime of Components	24 months
Lower bound replacement cost of VV + First Wall over lifetime of Plant (30 years)	\$4352/kW

fabrication methods, and as a result, the fabrication costs represent 93 % of the total costs associated with the vacuum vessel, blanket, and magnet systems [16]. The method used involved comparing the overall projected costs of four burning plasma designs, namely FIRE, BPX, PCAST5, and ARIES-RS, against the weight of each device from the cryostat inward [37]. The methods used in the ARC paper result in the cost of manufacturing being roughly \$1000 per kilogram, which more closely represents a FOAK fabrication cost than a NOAK fabrication cost. Thus, for this TEA, the \$1000/kg assumption will be used as an approximate FOAK cost. This section will demonstrate how a NOAK cost can be derived from this FOAK cost.

A variety of methods were employed to derive the lower and upper bound of the NOAK fabrication cost. To understand the timeline of the NOAK plant, the following assumption will be made: there will be a 100 GWe installed capacity by 2050, and at 100 GWe, there will be $\sim 300,350$ MWe FPPs [19]. There are several ways to take the FOAK cost and arrive at the lower bound fabrication cost of $\sim \$150/\text{kg}$. Firstly, by applying Wright's Law to the FOAK cost assumption (Fig. 5), by the 300th unit, fabrication cost per kg is in line with the lower bound assumption. To assess the full impact of a learning rate, the learning rate of solar technology, which follows Wright's Law, was considered; thus, the impact of applying a 20 % learning rate is shown in Fig. 5 [38]. The magnets are expected to be mass-produced in a factory-like environment, similar to Solar PV, and as such, the application of this learning rate is somewhat justified. However, applying this learning rate to the entire ARAI-FPP is not justifiable. In addition, for context, even highly modularized thermal power plants, like natural gas plants, experience learning of about 15 % [39].

Table 11
Applying a factor of 10 cost savings to Lower Bound Fabrication Cost.

DT Magnetic Confinement Reactor Base Case	Conventional fab. Cost (\$150/kg)	Advanced Manufactured Cost (\$15/kg)
Fabricated Magnet Cost	\$1371/kW	\$131/kW
Fabricated Blanket Tank + TiH ₂ Shield Cost	\$63/kW	\$8/kW
Fabricated VV + First Wall Cost	\$109/kW	\$30/kW
HTS Fabricated Costs	\$162/kW	\$24/kW
Lifetime of Replaceable Components	24 months	
Replacement of VV + First Wall over lifetime of Plant (over 30 years)	\$1632/kW	\$448/kW

Similarly, the lower bound cost of approximately \$150/kg for fabrication represents costs associated with conventional fabrication of low-alloy steel for NPP pressure vessel costs, inclusive of labor, factory equipment costs, quality assurance, and qualification [40,41]. Hence, \$150/kg was selected as the lower bound NOAK fabrication cost. Table 10 shows the cost of the fabricated power core components with this assumption.

4.4. Sensitivity on fabrication cost

Applying fabrication costs (assuming NOAK) based on conventional fabrication methods such as welding dramatically reduces fabrication costs from what is estimated in ARC [16]. However, two additional assumptions can be made to further reduce the existing fabrication costs lower bound of \$150/kg. 1) Apply the lower bound fabrication cost of mass-manufactured technologies to the lower bound fabrication costs of the TEA. 2) Assume that with advanced manufacturing or additive manufacturing, the fabrication cost will drop significantly. Both cases are assumed to reduce the cost of fabrication by a factor of 10 as discussed in this section.

The ARIES-ST fabrication study outlines a method of applying additive manufacturing techniques to the most critical component of the power core, which is also a replaceable component, the centerpost [42]. Unlike ARAI, the ARIES-ST has a large centerpost that will need to be replaced every 2–3 FPY. The lifetime of power core components differs based on reactor design. An example of this difference is related to the blanket in this ARIES-ST approach, where the blanket serves as a replaceable component. While the vacuum vessel in the ARIES-ST is designed to last the lifetime of the plant, the TEA assumes that the vacuum vessel and plasma-facing components are replaceable. Within the ARIES-ST studies, a low-cost fabrication approach and conventional fabrication approach were studied and applied to specific power core components and the costs and processes were compared. All costs reported pertaining to the ARIES-ST fabrication study will be inflated to \$2021 unless noted otherwise [42].

The ARIES-ST fabrication study approximated the centerpost to cost \$132/kg (compared with \$5/kg material cost) if fabricated by conventional means which is inclusive of the material cost. The study also detailed the cost to conventionally fabricate a vacuum vessel using welding, and the Unit Cost of finished mass (39,776 kg) was estimated to be \$160/kg. This is in the same order of magnitude as the lower bound assumption of fabrication in this study, \$150/kg. Assuming a low-cost fabrication method, they estimated a cost for the fabrication of the centerpost using the laser (or plasma Arc) advanced technology, making the unit cost approximately \$13.19/kg. This cost is based on scaling up the production of smaller components in a system that is assumed to be highly automated with minimal operator oversight. Additionally, the hardware costs for the laser advanced technology are not included in the \$/kg estimate, as they are expected to be accounted for as a one-time capital expenditure. Nonetheless, the costs of conventional and advanced technology approaches both assume NOAK assumptions that

Table 12
Estimate of parameters impacted by regulatory assumptions.

Parameters Sensitive to Regulatory Assumptions	Lower Bound	Regulatory Escalation on Lower Bound
Structures & Site Facilities	\$819/kW	\$1317/kW
Direct Costs (\$/kW)	\$6005/kW	\$6958/kW
Indirect Costs (\$/kW)	\$1146/kW	\$2644/kW
Annual Operation and Maintenance	\$35/MWh	\$73/MWh
	\$140/ MWh	
LCOE		\$266/MWh
Capacity Factor	0.7	0.5

reflect costs associated with mature design and NOAK fabrication processes. The study concluded that the cost of the centerpost would be “approximately ten times higher” if constructed using conventional fabrication approaches. The laser forming process, allowing a generous allowance for downtime resulted in a fabrication of 8 months or so. The study recommends the fabrication of the centerpost to be done on the reactor site because shipping would be challenging. If one were to take the conclusions of the low-cost approach to fabricating the centerpost and apply it to the other fabricated components, one would see a tremendous reduction in capital costs. For application of lowest fabrication cost to ARAI-FPP, it is reasonable to assume that the \$/kg of fabrication cannot be less than the cost of fabrication of solar PV panels or common mass-produced products such as cars, which are coincidentally about ~\$15/kg [43]. Reducing the reactor component fabrication costs by a factor of 10, from \$150 to \$15/kg would cause the FPP to reduce capital costs by roughly \$1200/kW over the plant’s lifetime. Table 11 depicts the costs after applying this factor, which will be labeled as “Advanced Manufactured Cost.” Future R&D, learning rates, and production path for the reactor equipment for ARAI-FPP will determine its cost potential. However, Wright’s law has limitations and must be applied cautiously by fusion industry business planners to relevant FOAK components.

There are other considerations that drive learning and capital cost considerations; for example, CF impacts how fast NOAK and learning can be achieved. With any new technology with high capital costs, achieving a high CF for the first several years of operation will be challenging, as demonstrated by the fission and carbon-capture fossil power plants as well as many solar thermal power plants and offshore wind. Slower deployment rates for these technologies lead to a slower decline in their FOAK costs. In addition, wind turbines, photovoltaic, and lithium-ion batteries have realized very steep learning rates, as assumed for the lower bound of this work, because their low overnight capital cost per unit lend themselves to rapid prototyping, R&D, and manufacturing at reasonable investment levels. Even at the lower bound of NOAK overnight capital cost estimate, \$3 billion is required to deploy a 350 MWe FPP.

4.5. Impact of regulatory framework on cost

Regulatory considerations are also relevant when understanding the sources of capital cost escalation. As noted in this analysis, the lower and upper bounds for the cost estimates mainly differ in the assumed regulatory oversight required for the construction and installation of FPPs. Table 12 depicts the impacts of the regulatory assumptions on the relevant cost accounts. The calculations were performed by applying the upper bound cost assumptions as outlined in previous sections, leaving the fabrication cost the same as the lower bound case [7]. The significant financial impacts induced by regulation can motivate the fusion industry to reduce its nuclear footprint as much as possible. Fusion energy systems do not pose the same risks as fission energy because the radioactive inventory in fusion is significantly lower than fission per power produced. Thus, major portions of the regulatory framework for fission energy are not germane to fusion. In addition to regulatory considerations, social acceptance can play a critical role in the deployment and

Table 13
Overnight Capital Cost, Annual Operations and Maintenance Cost, and LCOE.

	Lower Bound	Upper Bound
OCC	\$8831/kW	\$22,180/kW
Annual O&M	\$35/MWh	\$182/MWh
LCOE	\$140/MWh	\$550/MWh

cost of fusion energy projects. Drawing on lessons from the fission industry, where a lack of social acceptance has resulted in higher capital costs and regulatory burdens, the fusion industry should proactively engage the public to build trust and understanding [44]. By encouraging open dialogues about the benefits and risks, fusion can differentiate itself from fission, potentially easing regulatory paths and reducing costs.

In April 2023, the U.S. Nuclear Regulatory Commission (NRC) approved NRC's staff option to regulate fusion technology under 10 CFR part 30, "Rules of General Applicability to Domestic Licensing of Byproduct Material," which focuses on material that may pose a hazard [45]. The fusion industry has been supportive of this NRC decision. Note that this decision calls for a new volume of "Consolidated Guidance About Materials Licenses" dedicated to fusion energy systems and leaves open that additional regulations may be required if the "anticipated fusion design presents hazards sufficiently beyond those of near-term fusion technologies" [45]. However, the regulatory details of the NRC rulemaking are presently being worked on and are expected to evolve as industry stakeholders and regulatory bodies continue to collaborate and refine the guidelines to better align with the unique characteristics and safety profile of fusion energy technologies.

In July 2024, the U.S. Congress passed the Fusion Energy Act of 2024 as part of the bipartisan Fire Grants and Safety Act [46]. The legislation expands the Atomic Energy Act of 1954 by treating fusion machines separately from fission reactors and requires the NRC to develop a dedicated licensing scheme for fusion devices by 2027. The policy intent is to remove the classification of fusion within a governance regime structured for fission to create, more appropriately, a risk-based regulation scope commensurate with the safety and risk features of fusion.

While the focus of this TEA has been on D-T MC reactors, it should be noted that both D-T and non-D-T designs can realize reduced regulatory burdens compared to fission. The recent ruling by the U.S. NRC changed the past thinking that only non-D-T fusion could leverage regulatory benefits. If D-T FPPs maintain a low radioactive inventory by relying on low activation materials in the construction of FPPs and reduced tritium usage, they have the potential to be regulated under the same framework as non-D-T fusion. This reduced scope is expected to result in significantly ($>2\times$) lower costs of fusion-related equipment and civil structure, as shown by the sensitivity of the lower and upper bound costs for ARAI-FPP. All in all, the fusion industry should work to strike a balance amongst economies of scale, fusion fuel type, regulatory oversight, and engaging public support, which is instrumental in navigating the path to commercial technology.

5. Conclusion

Fusion technology currently hosts uncertainties that cast a speculative shadow over the results of this analysis. Nevertheless, this paper provided a unique capital cost methodology and comprehensive evaluation of NOAK technologies for an FPP while identifying the requisite fusion-specific FOAK technologies despite their inherent uncertainties.

The OCC and the annual O&M costs greatly impact the LCOE of an FPP. Meanwhile, the direct costs are the dominant cost of the OCC of a NOAK FPP. It was shown that the reactor plant equipment, which hosts the most fusion-specific technology, is the cost driver of the direct costs. A cost driver analysis of the reactor plant equipment revealed the significant impact of fabrication costs on power core components, thus suggesting that reducing costs by a factor of 10 could save at least

~1200 \$/kW (for the lower bound estimate) over the plant's lifetime. Additionally, this cost driver analysis and the assumptions regarding reactor plant equipment directly affect variable O&M costs, as these costs encompass the fabrication of replaceable components.

Moreover, given the large cost increase in fission, driven by regulatory oversight, provided the fusion industry succeeds in maintaining a minimal radioactive footprint and gains social acceptance, it is reasonable to anticipate that future regulatory frameworks will not impose severe cost penalties in the U.S. This harmony amongst regulation, cost management, and societal engagement is essential for rendering safe and economically viable fusion energy.

This analysis recognizes many sensitive parameters and acknowledges that the cost figures presented are the best estimates based on currently available information. The final outputs of this analysis are shown in Table 13, which only is applicable to the magnetic-confinement concept. The most effective method for refining the overnight capital cost estimates for FPPs would be for companies to share their cost data transparently. Despite this, the true cost of building an FPP will remain uncertain until the FOAK plant is constructed. Nonetheless, this analysis is a vital reference in understanding the economic landscape of fusion technology, informing future R&D and regulatory decisions.

CRedit authorship contribution statement

Layla S. Araiinejad: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Korosh Shirvan:** Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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